# Question 4

## Introduction

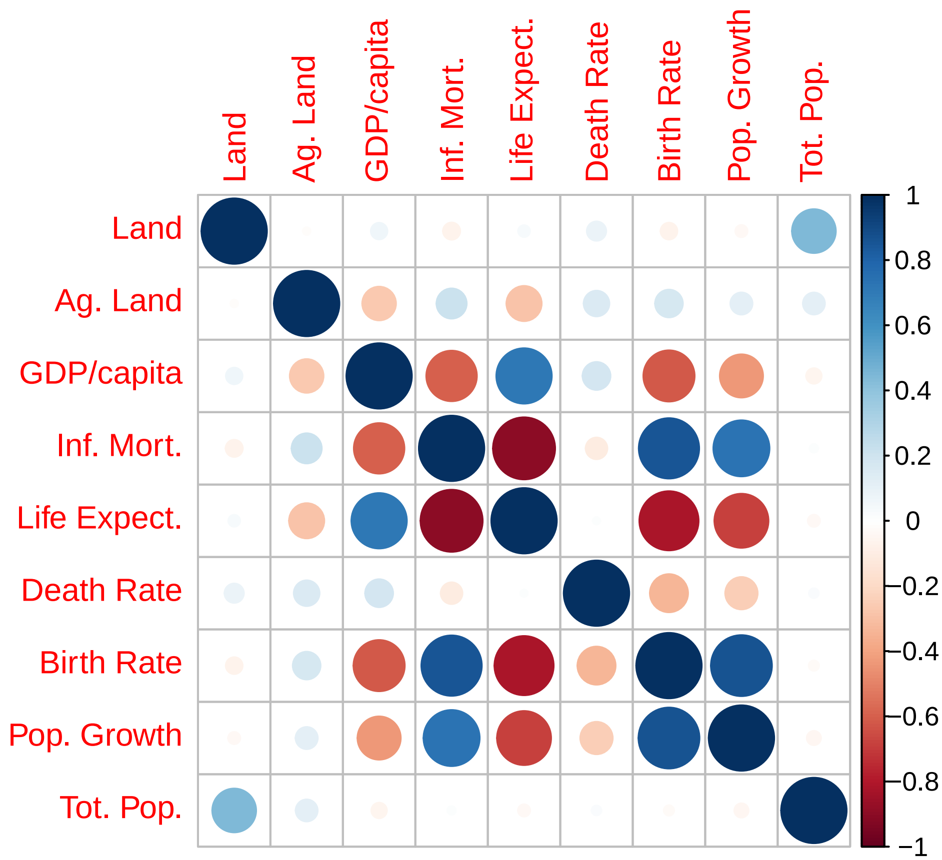
In this section, we focus on the question: "What do patterns in land use, demographics, and economics reveal about global inequality and regional development?"

We examine the variables: Total Land, Arable Land, Real GDP per capita USD, Infant Mortality Rate, Life Expectancy at Birth, Death Rate, Birth Rate, Population Growth Rate, and Total Population.

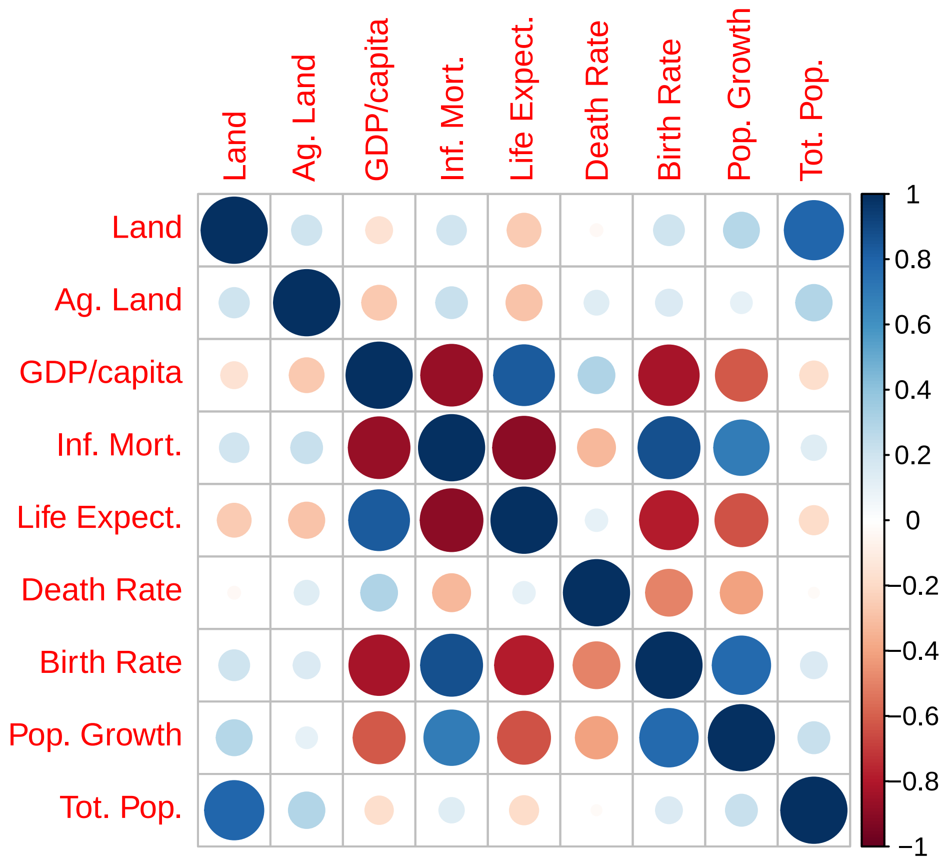
The techniques we use are: Corrplots, Biplots, Boxplots, and Scree Plots (Constituent of PCA).

## Correlation

In this section, we examine the correlation between numeric variables by means of a corrplot.



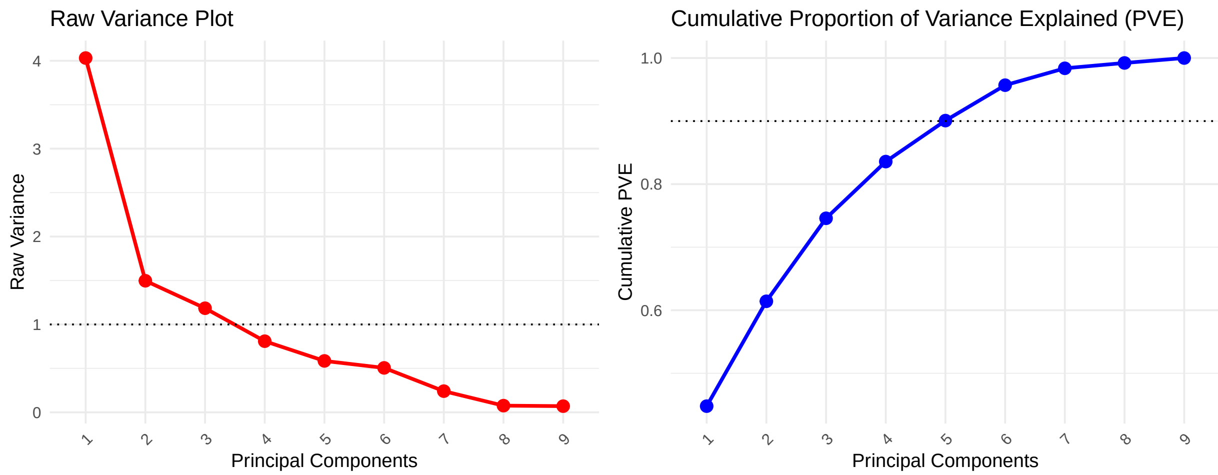

In this plot we see the Pearson correlation between the aforementioned numeric variables. Specifically, one can see that land, agricultural land, death rate, and total population have little correlation with other variables whereas the remaining variables seem to have complex correlation patterns among themselves. In this plot, one sees a strong negative correlation between life expectancy and infant mortality, a strong positive correlation between birth rate and population growth, a strong positive correlation between birth rate and infant mortality rate, and a strong negative correlation between birth rate and life expectancy. These correlations are linear and could perhaps be dependent on an underlying phenomenon such as access to medical care.



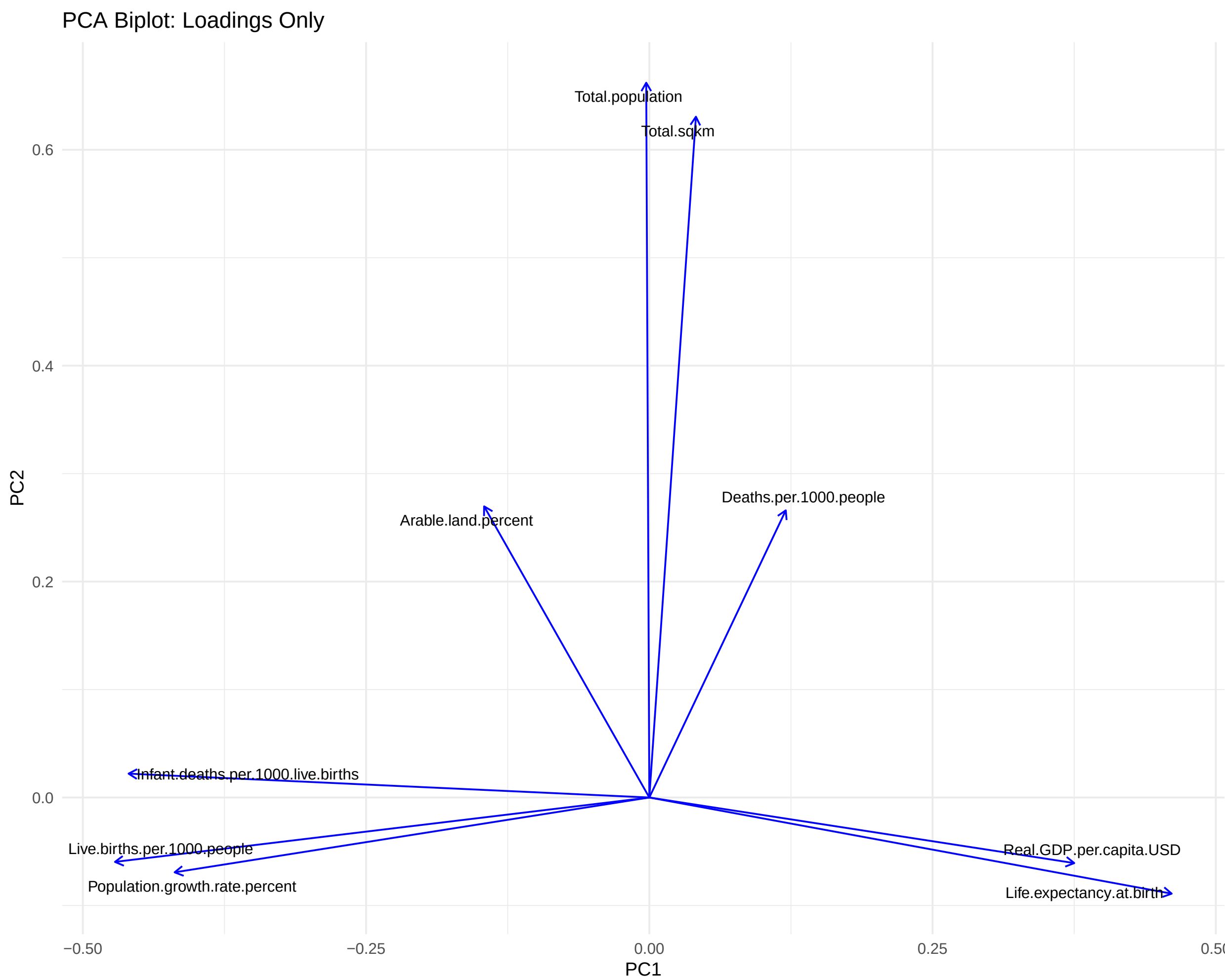

In this plot we see the Spearman correlation between the numeric variables. Although similar to the above plot, one can see higher positive correlation between land area and total population, suggesting that there is a monotonic relationship between the two variables though perhaps nonlinear. One can also see a stronger negative correlation between infant mortality and real GDP per captia, again suggesting that there is a strong, but non-linear relationship. There appears to be a weaker positive correlation between birth rate and population growth, perhaps suggesting that this is a linear relationship.

## Principal Components Analysis

In this section we do principal components analysis on the same variables as in `Correlation`. We begin the analysis by looking at the variance contained in each of the principal components, then we examine the principal component loading vectors and formulate interpretations of each of the first two principal components based on the loadings of each original variable. Finally, we look at the scores of the observations and examine per-continent trends based on our interpretation of the first two principal components.




There are 3 principal components that "aggregate information" due to the variance being larger than 1 while the PCA was run on scaled data. Additionally, to capture ~90% of the variance, we need 5 PCs, and to capture ~75% of the variance, we need only 3 PCs.

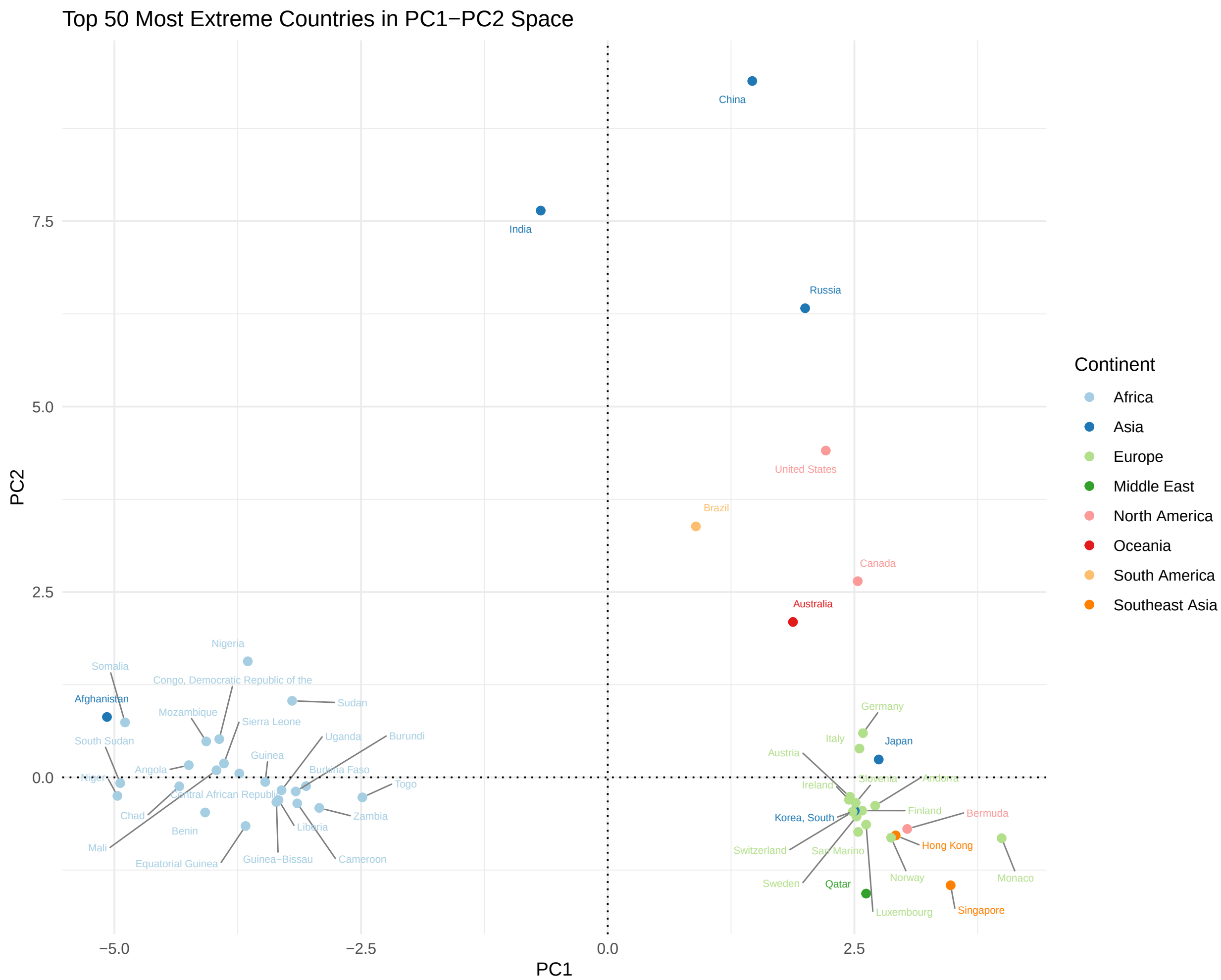



In the above biplot of the principal component loading vectors we make the following observations:

One can see that for small values of PC1 we see large values of Infant Mortality Rate, Birth Rate, and

Population Growth Rate. For large values of PC1 we see large values of Real GDP per capita USD and Life Expectancy at Birth. One could interpret PC1 as the developedness of countries with larger values corresponding to more developed countries.

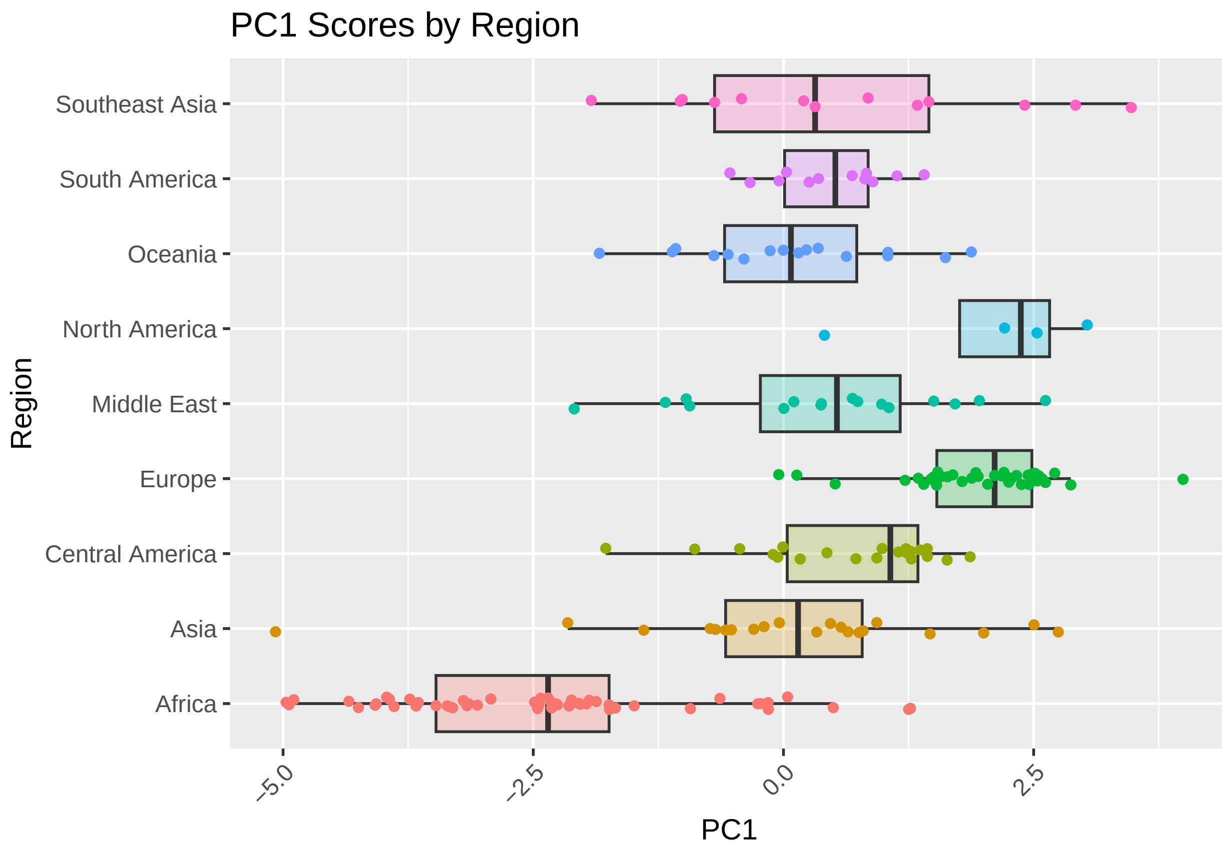
For large values of PC2 we see that the countries typically have large values of Total Poplation and Total SqKm. To a lesser degree Arable Land and Death Rate also contribute to the PC2 direction. An interpretation of PC2 could be the size of a country in terms of population and area.




In the scores plot we plot the scores of the 50 countries farthest from the origin. We also make the following observations and conclusions:

We see groupings along the PC1 axis of more developed countries on the side with larger PC1 values and less developed countries with smaller PC1 values. There are several outliers with large values for PC2 that correspond with larger countries in terms of both population and land mass. Specifically, countries from Africa tend to have lower values of PC1 whereas countries from North America, Europe, and Southeast Asia tend to have larger values for PC1. Additionally, it appears that strong outliers account for the majority of the variance in PC2 as we see India, China, Russia, and the United States with the largest values whereas most other countries group near 0 with less variation.

### Examination of PC's by Continent




In these boxplots, one can see that Africa tends to have smaller values for the PC1 with the lowest median. This indicates less developedness by our earlier interpretation. North America has the highest median. The country with the highest value for PC1 is from Europe (Monaco, from above) and the country with the lowest value is from Asia (Afghanistan, from above).

## Conclusion

Through this analysis, we sought to answer the question: "What do patterns in land use, demographics, and economics reveal about global inequality and regional development?" To do so, we examined linear and monotonic correlation between the variables themselves with Corrplots and continent/country-specific information, aggregated with PCA.

The main takeaways are as follows: The variables we examined tend to cluster into a few groups such as "size" of countries, "wealth" of countries, and "poorness" of countries. There are only a few "large" countries in terms of land mass and population, namely India, China, Russia, and the United States. Based on the variables examined and our interpretation of the Principal Components, Africa tends to be less developed than the others. From this analysis and the geographic grouping seen in the anaylsis, one could speculate that the geographic location (among other factors such as colonialization) plays a large role in the developedness of nations.